

Soln: a) Suppose all the inscriptions are false.

Trunk 1 says: "The treasure is in Trunk 3"
 $\therefore$ The treasure is in either Trunk 1 or Trunk 2

Trunk 2 says: "The treasure is in Trunk 1"
 $\therefore$ The treasure is in Trunk 2

Trunk 3 says: "This trunk is empty" which is True. This is a contradiction to the fact that ~~the~~ "all the inscriptions are False" being True
 $\therefore$ The Queen cannot say this.

b) Let, exactly one of the inscriptions be True.

Let Trunk 1 be True. $\therefore$ The treasure is in Trunk 3
Trunk 2 states that the Treasure is in Trunk 1, which is false and consistent with the fact that exactly one of the inscriptions is True.

Trunk 3 states that it is empty which is False and consistent.

Let Trunk 2 be True. $\therefore$ The treasure is in Trunk 1
Trunk 1 states that the Treasure is in Trunk 3 which is False and consistent.

Trunk 3 states that it is empty, which is True. But it must be False, otherwise it's a contradiction to the fact that exactly one of the inscriptions is True. $\therefore$ Trunk 2 must be False

Let Trunk 3 be True. $\therefore$ It is empty.

Since Trunk 2 must be False, the treasure is not in Trunk 1. So, it must be in Trunk 2.

Trunk 1 states that the treasure is in Trunk 3 which is false and consistent.

$\therefore$ The Queen can say this, but the treasure is in either Trunk 2 or Trunk 3

c) Let, exactly two of the inscriptions be True.

Let Trunk 1 be False. The treasure is not in Trunk 3.
Since Trunk 2 is True, the treasure is in Trunk 1

Trunk 3 is stating that it is empty which is True and consistent.